



Do Macroalgal Mats Impact Microphytobenthos on Mudflats? Evidence from a Meta-Analysis, Comparative Survey, and Large-Scale Manipulation

Alice F. Besterman¹ · Michael L. Pace¹

Received: 8 December 2017 / Revised: 16 April 2018 / Accepted: 12 May 2018 / Published online: 30 May 2018
© Coastal and Estuarine Research Federation 2018

Abstract

Eutrophication and species introductions have resulted in increased macroalgal biomass in coastal ecosystems around the globe. Macroalgal mats may compete with microphytobenthos (MPB) for light and nutrients and, due to their position in the canopy, have a negative impact on MPB biomass. We tested this effect by conducting a meta-analysis of prior experiments, as well as a comparative survey, and a macroalgal-removal manipulation in the coastal lagoons of the Virginia Coastal Reserve (VCR) on the eastern shore of Virginia (USA). In all cases, MPB biomass was estimated using benthic chlorophyll as a proxy. While prior individual studies documented impacts of macroalgae, when effect sizes were averaged across studies, there was no consistent effect of macroalgal biomass on MPB biomass. In the VCR, a non-native red macroalga, *Gracilaria vermiculophylla*, dominates intertidal mats and attains high biomasses at some sites. Nevertheless, MPB biomass was unrelated to macroalgal mass based on a survey of mudflats. Further, when macroalgae were experimentally manipulated on a mudflat using a before and after impact design, there was no change in MPB. Based on the meta-analysis, survey, and manipulation we conducted, macroalgal mats do not have a generalizable effect on MPB, interactions seem context-dependent, and in the VCR, the effects on MPB appear neutral. This finding is important given the significance of MPB in supporting food webs and other estuarine ecosystem functions, as well as the increasing frequency and intensity of macroalgal blooms.

Keywords Competition · Microphytobenthos · Macroalgae · Invasive · *Gracilaria vermiculophylla* · Diatoms

Introduction

Seaweeds have proliferated globally in shallow coastal waters as a consequence of species introductions and eutrophication (McGlathery et al. 2007; Williams and Smith 2007; Williams et al. 2013; Lyons et al. 2014). Seaweeds can accumulate into thick, contiguous blankets that cover large areas. A diverse community of macroalgal taxa can be found at both subtidal and intertidal elevations in coastal bays, and many species can remain healthy on tidal flats despite experiencing extreme

temperatures and temporary desiccation (Thomsen et al. 2006; Thomsen and McGlathery 2007). Macroalgae have the potential to negatively impact the abundance and productivity of other autotrophs in these systems such as seagrass and microphytobenthos (McGlathery et al. 2001, 2007, 2012; Lyons et al. 2014). The negative effects of macroalgal blooms on seagrass meadows are well understood, but the interactions between microalgal biofilms and macroalgae remain uncertain (Valiela et al. 1997; Sundbäck and McGlathery 2005; McGlathery et al. 2007). Microphytobenthos (MPB) are microscopic algae such as diatoms that form biofilms on benthic surfaces. MPB are significant estuarine primary producers, support estuarine food webs, stabilize sediments, and regulate nutrient fluxes between sediments and water (Cahoon 1999 and references therein; Weerman et al. 2010; Douglass et al. 2011; McGlathery et al. 2012). Up to 99% of total benthic primary production in estuaries may be accounted for by MPB, and rates are often equal to or greater than water column production (Nilsson et al. 1991, Underwood and Kromkamp 1999, McGlathery et al. 2001). Microalgae are directly consumed by benthic invertebrates, as well as some shorebirds

Communicated by Rui Santos

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s12237-018-0418-3>) contains supplementary material, which is available to authorized users.

✉ Alice F. Besterman
afb5kg@virginia.edu

¹ Department of Environmental Sciences, University of Virginia, 291 McCormick Rd., Clark Hall, Charlottesville, VA 22903, USA

and fish (Cahoon and references therein 1999; Douglass et al. 2011; Kuwae et al. 2012; McGlathery et al. 2012; Carpentier et al. 2014). Consequently, benthic microalgae support significant secondary production in estuaries (Currin et al. 2011; Douglass et al. 2011; Saint-Béat et al. 2013). MPB also stabilize sediments by the production of binding extra polymeric substances (EPS); these colloidal carbohydrates function as a gel which increases erosion thresholds (Cahoon 1999 and references therein; Weerman et al. 2010; Mariotti and Fagherazzi 2012; McGlathery et al. 2012). Because fluxes of remineralized nutrients from sediments to surface waters are limited by MPB uptake, MPB are critical in regulating nutrient dynamics and minimizing the likelihood of harmful phytoplankton blooms (Cahoon 1999 and references therein; McGlathery et al. 2001; Tyler et al. 2003; Sundbäck and McGlathery 2005). Decreased microalgal biomass could diminish or eliminate these ecological functions, resulting in altered food webs, higher turbidity, and higher nutrient concentrations in estuarine waters.

Macroalgal thalli and MPB interact in complex, non-linear ways as they compete for light and nutrients (Sundbäck and McGlathery 2005; Hardison et al. 2013). For instance, thick macroalgal mats attenuate light rapidly; irradiance may be reduced by 90% within a few centimeters (Krause-Jensen et al. 1996; Sundbäck and McGlathery 2005). The reduction in light intensity would theoretically induce competition, whereby the macroalgal canopy physically blocks MPB from receiving light. While light is clearly a limiting factor to primary producers, diatoms (the dominant group of microalgae in benthic mats) are capable of adapting to low light environments by increasing pigment production and light use efficiency (Cahoon 1999 and references therein; Sundbäck and McGlathery 2005; García-Robledo et al. 2012). Microalgae can also somewhat tolerate anoxia and sulfides, both of which may result from macroalgal mats decaying at the sediment-water interface (Cahoon 1999 and references therein; Kennett and Hargraves 1985). Competition between macro- and microalgae is also possible in nutrient limited systems. Macroalgal mats could utilize available nutrients in the water column more rapidly than MPB, limiting microalgal production (Sundbäck and McGlathery 2005). Despite the high tolerance of microalgae to these competitive effects, there may be some thresholds beyond which microalgae are negatively affected by macroalgae.

Gracilaria vermiculophylla (hereafter *G. vermiculophylla*) is a non-native, red macroalga that has flourished along the Atlantic Coast of the USA since a cryptic introduction occurring at least 30 years ago (Thomsen et al. 2005; Gulbransen et al. 2012). This alga maintains significant population densities throughout the year and achieves high biomasses at some locations (McGlathery et al. 2001; Thomsen et al. 2006). In the coastal bays along the eastern shore of Virginia, *G. vermiculophylla* is the dominant species of macroalgae; thalli

are prolific in intertidal and subtidal habitats (Thomsen et al. 2006; Gulbransen et al. 2012, pers. obs.) (Fig. 1). Macroalgae in Virginia, especially *G. vermiculophylla*, can be incorporated into the tubes of the native polychaete *Diopatra cuprea* (Thomsen and McGlathery 2005b). The majority of *G. vermiculophylla* thalli found in Virginia's seaside bays are stabilized near the sediment by these worm tubes (Thomsen and McGlathery 2005b; Thomsen et al. 2009). This facilitative interaction has also been found in Georgia (Kollars et al. 2016). While the abundance, distribution, and effects of *G. vermiculophylla* have been considered, the effects on MPB have mainly been studied in mesocosms (Thomsen et al. 2006; Thomsen and McGlathery 2007; Hardison et al. 2010; Hardison et al. 2013). In the Virginia Coast Reserve (VCR), a network of protected coastal habitats, there is significant variability in biomass of macroalgal mats among intertidal areas with possible but unknown consequences to benthic microalgal biomass beneath the mats.

Given trends causing proliferation and high biomasses of macroalgae, we hypothesized a negative impact of macroalgae on MPB. In this study, we tested this idea using three approaches; in each, we used benthic chlorophyll-*a* (hereafter benthic chlorophyll) as a proxy for MPB biomass. We summarized effect sizes from prior experiments that tested macroalgal-MPB interactions with a variety of macroalgal species from a variety of sites. We conducted a macroalgal



Fig. 1 Image of macroalgal mat near the macroalgal removal manipulation site during June 2015. This is a heavily vegetated mudflat for the Virginia Coast Reserve (VCR), but typical for the mudflats around Oyster Harbor. The brown-red colored surfaces are non-native *Gracilaria vermiculophylla* and the green patches are primarily foliose *Ulva* spp. (color figure online)

survey to estimate average biomass for the VCR and also measured macroalgal biomass in relation to MPB biomass to test for their relationship. Finally, using a before and after impact experimental design, we removed macroalgae from a mudflat and compared the response of MPB with an adjacent reference mudflat.

Methods

Meta-Analysis

We used database searches, paper bibliographies, and consulted papers we were aware of to locate studies on the interactions of macroalgae and MPB. Keywords searched for in ISI Web of Knowledge database included various combinations of macroalg*, microalg*, microphytobenthos, interaction*, benthic, and shading. Papers were included if benthic chlorophyll results were reported as a measure of MPB biomass. Information was extracted from all independent treatments from each study. This resulted in multiple samples for the meta-analysis from each of the studies. Some studies were designed as repeated measures experiments where plots or mesocosms were sampled over time. The meta-analysis included studies of both intertidal mudflats and shallow subtidal sediments. Data used in the meta-analysis were either explicitly published in the text or tables of studies or obtained from figures. Effect sizes were calculated from experiments by subtracting the mean concentration of benthic chlorophyll in the treatment (with macroalgae) from the control (without macroalgae). A measure of effect size that accounts for standard deviation, Cohen's d , was also calculated (Lee 2016). The mean effect size and mean Cohen's d across all studies were calculated. While there was some non-independence of samples due to repeated measures designs in studies used, and multiple treatments extracted from single studies, we did not use a nested or repeated measures design for assessing effect size or Cohen's d . Mass reported as wet weight was transformed to dry weight using the relationships in Sutula et al. (2014). Effect size was also compared with macroalgal mass used in the treatment. In order to determine whether there was an effect of macroalgae on MPB in the meta-analysis, 95% confidence intervals for mean effect size or Cohen's d were tested for overlap with zero (Lee 2016).

Study Sites

The field research was done at the Virginia Coast Reserve, which is a 110-km-long (north to south) protected region of coastal habitats including marshes, mudflats, lagoons, tidal creeks, and barrier islands on the seaside (eastern coast) of the Delmarva Peninsula (Fig. 2). The distance between the mainland and barrier islands varies significantly, from 14 km

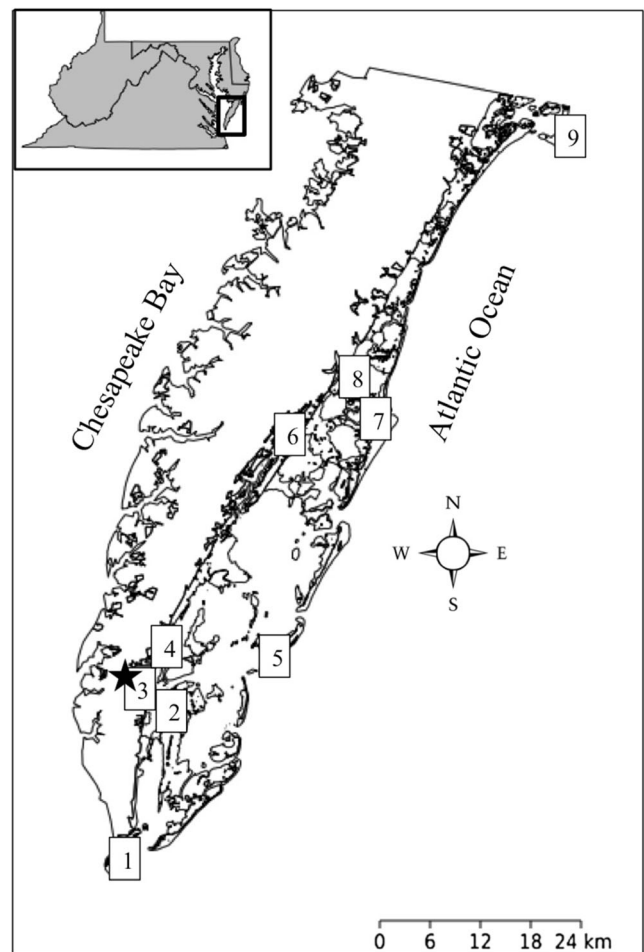


Fig. 2 Map of the Virginia portion of the Delmarva Peninsula. Inset map shows the US mid-Atlantic coast with box around the Virginia Coastal Reserve (VCR). The VCR includes the majority of the peninsula depicted (excluding the most northern portion, location of site 9). The nine study sites used for the macroalgal survey in 2015 are numbered as follows: (1) Fisherman's Island (37.106049°, -75.962281°), (2) narrows (37.266702°, -75.904427°), (3) Oyster Harbor (37.286244°, -75.914393°), (4) Coast Guard (37.293724°, -75.918173°), (5) Cobb Island (37.309754°, -75.772435°), (6) Quinby (37.566211°, -75.746768°), (7) Curlew Bay (37.591826°, -75.658526°), (8) Hummock Cove (37.615114°, -75.662684°), (9) Tom's Cove (37.886435°, -75.348119°)

at the widest point to ~2 km at the narrowest. Land use on the adjacent mainland is mainly agricultural; however, nutrient concentrations in VCR coastal lagoons are low (McGlathery et al. 2001, 2007). Sites selected for the macroalgal survey spanned the entire north-south extent of the VCR as well as sites along the east-west axis from the mainland to lagoons and barrier islands (Fig. 2). One site was just beyond the northern boundary of the VCR and is more impacted by anthropogenic developments as the barrier island is part of a national park, and the beach is open to public use (Tom's Cove on Assateague Island; Fig. 2). Mudflats varied in area, elevation, and microtopography due to the presence of channels, oyster reefs, and other features representative of the

VCR. These locations were chosen to represent a range in total macroalgal biomass and *Gracilaria vermiculophylla* dominance. Macroalgal biomass and diversity peak at mid-lagoon locations along the east-west axis (Thomsen et al. 2006). Nutrients also may be higher in the more northern bays of the VCR, and we consequently would expect macroalgal biomass to increase from south to north (unpublished data).

We conducted a comparative survey on mudflats along the lagoons and channels near Oyster, Virginia (Fig. 2). Twenty-two sample locations were haphazardly selected; all sites were at least 20 m apart and were near to the Narrows, Coast Guard, and Oyster Harbor sites (Fig. 2). The barrier islands are about 11 km east of Oyster, and sites were all concentrated within 2.5 km of the mainland and within a $\sim 9 \text{ km}^2$ area. We wanted to reduce the variability of any environmental parameter other than macroalgal biomass. By sampling within a relatively small area, we minimized effects of turbidity, nutrients, and hard substrate availability, which all have been found to vary between the mainland and barrier islands and affect macroalgal and MPB biomass (Thomsen et al. 2006).

For the large-scale macroalgal manipulation, the Oyster Harbor site sampled in 2015 was chosen. There is extensive intertidal mudflat cover of macroalgae, and *G. vermiculophylla* in particular, in that area (Fig. 1). The significant amount of *G. vermiculophylla* and total macroalgal biomass provided an opportunity to test for an interaction where this species dominated and where total biomass was high for the VCR. This site is also easily accessible by foot, which enabled us to sample more frequently than would have been possible if a site only accessible by boat was chosen.

Macroalgal Surveys and Comparative Survey

In May and July of 2015, benthic macroalgae biomass was measured throughout the VCR to assess variation on intertidal mudflats, especially of *Gracilaria vermiculophylla*. A total of eight mudflats were surveyed in May and nine in July (the same eight from May with one additional site added). We wanted to capture a representative biomass for macroalgae during summer in Virginia, when biomass is greatest during the annual cycle. Macroalgal biomass peaks around June and begins to decrease due to senescence in late July and early August (McGlathery et al. 2001; Thomsen et al. 2006). To represent this season adequately, we sampled near the beginning in May and the end in July. At each site, four 100 m transects perpendicular to the marsh edge were traversed with sampling points distributed every 10 m (transects were shorter at some sites because the mudflats were less than 100 m wide). At each sampling point, a 0.25-m^2 quadrat was used to quantify algal cover. All algae within one quadrant (0.0625 m^2) were collected into a plastic bag to measure biomass. Algal biomass samples were brought back to the laboratory and washed free of sediments and epifauna, then dried at 60°C , and weighed. Macroalgae were

separated into functional-form groups including: green foliose, green filamentous, red filamentous, and red coarsely branched; *G. vermiculophylla* was measured separately (McGlathery et al. 2012). Since macroalgal diversity was not a primary goal of this investigation, and non-*G. vermiculophylla* macroalgae were predominately *Ulva* spp., data were pooled into *G. vermiculophylla* and non-*G. vermiculophylla* groups. Macroalgal biomass means were computed as well as coefficient of variation for each site.

In September 2017, a comparative survey of mudflats was conducted along a continuum of macroalgal mass to test how MPB biomass varied along that gradient. Benthic chlorophyll was measured using spectrophotometric methods as a proxy for MPB biomass, a frequently used method (Lorenzen 1967; Cahoon 1999). Patches of macroalgae were selected to represent a wide range in mass. Twenty-two samples were collected from a 900-ha area of the VCR near Oyster Harbor. At each location, a 0.0625-m^2 quadrat was placed on the algal mat and three 2-cm deep cores were collected from the sediment directly beneath the macroalgae using a modified 5-cm^3 syringe. We did not observe macrophyte fragments (i.e., *Spartina*, *Zostera*, macroalgae) in these sediment cores that might affect measurements of chlorophyll of MPB. However, it is possible that some smaller macrophyte-derived particles influenced the variability of our benthic chlorophyll measurements. Sediment plugs were deposited into 15-mL centrifuge tubes and placed on ice immediately. All macroalgae were then collected into a plastic bag from within the quadrat and placed on ice. Centrifuge tubes were wrapped in tin foil to prevent chlorophyll degradation by light exposure, and these samples were frozen upon returning to the lab. A simple linear regression was used to test if macroalgal biomass was related to MPB.

Macroalga-Removal Manipulation

In 2016, a Before-After-Control-Impact experimental design was used to test if the removal of macroalgae from a heavily vegetated mudflat would result in an increase in MPB biomass (Smith 2002). This study design involved monitoring two large areas of mudflat over time. One area was left undisturbed as a control with macroalgal mat (control plot), and one area was manipulated halfway through the observational period by removing macroalgae (treatment plot). Thus, in this experiment, there were four treatments: before-control, after-control, before-treatment, and after-treatment. A semi-enclosed 140-m-long and 50-m-wide mudflat contiguous with the mainland near Oyster Harbor was selected for the removal experiment. The flat is bordered by saltmarsh on three sides and in most low tides becomes completely exposed. Macroalgal cover and biomass were nearly homogeneous, and tidal/current flows were similar. Half of the mudflat was designated as the “control” and the other half as the “treatment” site, each area being $70 \times 50 \text{ m}$. Later in the experimental period, a nearby naturally

bare mudflat was selected for qualitative comparison. In order to minimize disturbance to the macroalgal biomass before the manipulation, macroalgal cover was sampled and then converted into grams dry weight per square meter using data collected in 2015 (Table 1; Supplemental Material, Fig. 1A).

Regular sampling of MPB began on July 10 and continued for 30 days through August 9. Percent cover of macroalgae was measured before the removal on each plot along a single 50-m transect with points taken every ten meters using a 0.25-m² quadrat. The algae removal from the treatment site took place halfway through sampling period on July 22. Macroalgae were pulled from the flat by hand and using rakes, deposited in plastic sleds, and transported off the mudflat. Cover was assessed again after the removal in similar fashion as before except two 50-m transects were surveyed on the treatment plot to gain better resolution as the removal left the remaining algae quite patchy. The removal disturbance was not explicitly repeated on the control plot, but all evidence of the removal was gone after 2 days. Sampling resumed after 5 days to give the sediment and MPB sufficient time to

recover from the disturbance. MPB turnover rapidly and can respond to manipulations within a few days (e.g., Pace et al. 1979). Additionally, trampling has not been found to significantly impact MPB (Rossi et al. 2007; Leite et al. 2012). Macroalgae did not recolonize the treatment site after removal during the study.

Replicate samples of benthic chlorophyll were collected six times before the removal and six times after. Sampling always occurred at mid-tide (0.6–1.2-m water depth) to minimize disturbance to the sediment and to standardize the light and hydrologic variables for MPB sampling. We collected samples by walking on the site, but the slight positive buoyancy provided by the water over the mudflat helped to decrease the disturbance to the sediment. Sediment cores for chlorophyll were collected with modified 5-cm³ syringes to 1-cm depth. While MPB vertically migrate downward during flood tide, the bulk of the biomass is still within the top centimeters of sediment (Du et al. 2010). Generally, eight replicates were collected for benthic chlorophyll for each sampling event but on a few occasions replication was lower. Eight replicate cores

Table 1 Summary of prior studies of macroalgae (MA)-microphytobenthos (MPB) interactions when benthic chlorophyll (chl-a) was measured

Reference (year)	Experiment/study description	Results summary	
García-Robledo et al. (2012)	Indoor flow-through cores; foliose <i>Ulva</i> ; 3-week incubation	Chl-a in control and maximum MA level similar and lower than intermediate treatments	MA ↑: MPB ~
García-Robledo and Corzo (2011)	Indoor flow-through mesocosms; filamentous <i>Ulva</i> ; 14–20-day incubations	Decrease in chl-a under macroalgal mat	MA ↑: MPB ↓
Sundbäck et al. (1996)	Outdoor flow-through cores; filamentous <i>Ulva</i> ; 8, 11-, 14-, 21-, and 41-day incubations	No adverse effects from macroalgae on microalgae	MA ↑: MPB --
Sundbäck et al. (1990)	Outdoor flow-through cores; filamentous red MA; 1-, 9-, 16-, and 23-day incubations	No significant differences between treatments after 1, 9 or 16 days (only 16 days reported here). Control chl-a was significantly higher than low or high macroalgal levels on day 23	MA ↑: MPB ↓
Green and Fong (2016)	Field-intertidal mudflat, 1 m ² plots; <i>Ulva intestinalis</i> ; 2-, 4-, 6-, and 8-week incubations	No effect of treatment on chl-a; all treatments increased over time	MA ↑: MPB --
Bohórquez et al. (2013)	Field-intertidal mudflat, 2.25 m ² plots; filamentous <i>Ulva</i> ; one measurement per season	The presence of macroalgae increased concentrations of chl-a	MA ↑: MPB ↑
Corzo et al. (2009)	Field-intertidal mudflat, 2.25 m ² plots; filamentous <i>Ulva</i> ; 2–3-week incubations	Chl-a was lower in the control treatment than beneath macroalgae	MA ↑: MPB ↑
Hardison et al. (2013)	Outdoor flow-through mesocosms; <i>Gracilaria</i> spp.; initial conditions (pre-MA additions); 7, 14, 21, 29, and 42 days	No significant difference between macroalgal covered sediment and bare sediment mesocosms. Means for MA treatment were consistently lower, but very high variability	MA ↑: MPB --
Hardison et al. (2011)	Outdoor flow-through mesocosms with surface water nutrient enrichment; <i>Gracilaria</i> spp.; 7, 14, 21, 29, and 42 days	No significant difference between macroalgal covered sediment and bare sediment mesocosms. Means for MA treatment were consistently lower, but very high variability. Other variables collected indicate MPB productivity limited due to shading	MA ↑: MPB --
This study	Field, large-scale (> 3000 m ²); <i>Gracilaria vermiculophylla</i> and <i>Ulva</i> spp.	No significant difference between macroalgal covered and bare sediment; removal of MA did not increase chl-a	MA ↑: MPB --

Eight of these studies were included in the meta-analysis; the original results from this study were excluded, and Corzo et al. (2009) was excluded because the authors did not report quantitative results. Responses are depicted in symbolic notation as follows: macroalgae additions (MA ↑), non-linear pattern (~), no effect (--), positive effect (↑), and negative effect (↓)

are the minimum needed to estimate benthic chlorophyll at the meter-squared scale (Grinham et al. 2007). Locations to collect cores were chosen haphazardly from the plot. Before the manipulation, samples were collected beneath macroalgal mats on both plots, which were detected by feel. After the removal, samples on the treatment plot were collected from exposed mud, even though some macroalgae remained on the mudflat. Cores for benthic chlorophyll were placed in 15-mL centrifuge tubes. All samples were stored on ice until they could be transferred to a freezer at the laboratory. The removal manipulation was tested with a modified repeated measures ANOVA (Smith 2002). Using this technique, the intervention is considered significant if an interaction between the two factors exists: plot (control or treatment) and time period (before or after removal).

Key environmental variables for the mudflat were quantified concurrently with the MPB sampling. Spatial replicate samples were aggregated across the whole study period for these environmental variables. Sediment organic matter and porosity, total suspended solids (TSS), light intensity, salinity, and water temperature were measured. Sediment cores to measure organic matter and porosity were collected using a modified 60 cm³ syringe; five samples per plot were collected on each sampling occasion. Four 1-L water samples were collected on three occasions to estimate TSS from about 30 cm below the water surface. Salinity was measured with a YSI-probe on five sampling days. Temperature and light intensity were measured using three HOBO data loggers suspended 20 cm above the sediment. Macroalgal thalli remain attached to the sediment but float toward an erect position when the flat is flooded, elevating the top of the macroalgal canopy to 10–20 cm above the substrate. The loggers were positioned just above this canopy so incident light at the top of the submerged macroalgal mat was recorded. The loggers measured each variable every 10 min for the entirety of the experiment. We report averages for data collected during the hour concurrent with MPB sampling. This resulted in 11 samples each of light and temperature (measurements were compromised 1 day by algae fouling the loggers). Light data were transformed from Lux (the units used by HOBO dataloggers) to photosynthetic photon flux in micromoles per square meter per second using a standard conversion factor for sunlight of 0.0185 (Thimijan and Heins 1983).

Chlorophyll and Sediment Analyses

Frozen sediments in centrifuge tubes were thawed and protected from light exposure. Eight milliliters of a 45% methanol/45% acetone mixture was added to each test tube, and then samples were shaken to mix. Samples were then sonicated for 1 min to lyse cells and stored in a freezer for 24 h. After 24 h of extraction, tubes were centrifuged for 5 min. Supernatant was removed with a pipette and placed into cuvettes for analysis in a

spectrophotometer. Samples were measured at 665 and 750 nm, acidified with 5% HCL, and then measured again at 665 and 750 nm. Benthic chlorophyll concentrations were calculated using the equation from Lorenzen (1967).

Sediment organic matter was quantified using the loss-on-ignition method and presented as a percent of dry weight. Samples were first dried for at least 48 h at 60 °C. Then, samples were combusted in a muffle furnace. We estimated porosity, which is the percent of pore space in the sediment by volume. It was computed by dividing the mass of water lost during drying first by 1.025 g cm⁻³ (density of seawater) and then by the initial core sample volume of sediment and pore water.

Results

Meta-Analysis

Nine studies were identified that evaluated differences in MPB biomass beneath macroalgal mats (Table 1). However, one of these studies only reported a general response of MPB to macroalgal mats and did not provide a quantitative response (Corzo et al. 2009). This study is presented in Table 1 but excluded from the meta-analysis and Supplementary material, Table 1A, leaving eight total studies for evaluation in the meta-analysis. We were surprised that there were so few studies that measured both biomass of macroalgae and MPB; many studies examined the interaction of macroalgae and MPB but focused on other parameters such as productivity, nutrient fluxes, or oxygen fluxes. In all but one study, macroalgal biomasses were experimentally manipulated. The non-manipulative study monitored plots over a year by taking measurements from bare and macroalgal covered plots one time per season (Bohórquez et al. 2013). Enclosures were deployed to prevent macroalgae moving in or out, but the researchers did not control biomasses. Many of the experimental studies used varied levels of macroalgal mass and length of macroalgal incubation as treatment conditions. Three of the studies were field-based, plot-size experiments (maximum area 2.25 m²), and six were conducted in outdoor flow-through mesocosms or seawater-filled cores in a laboratory. Six studies used *Ulva* spp., two used *Gracilaria* spp., and one used a mixture of filamentous red macroalgae species. Effect sizes were calculated for each biomass level tested (most studies had multiple treatment levels), and Cohen's *d* was calculated for each treatment where standard deviation was provided, resulting in 37 and 35 values respectively (Supplementary Material, Table 1A).

MPB responses varied. Positive, negative, and non-linear responses occurred, and in a few cases, no effect was detected (Table 1; Fig. 3). While most of the macroalgal biomasses used in these experiments were significantly higher than the masses we measured in the VCR, the concentrations of MPB were

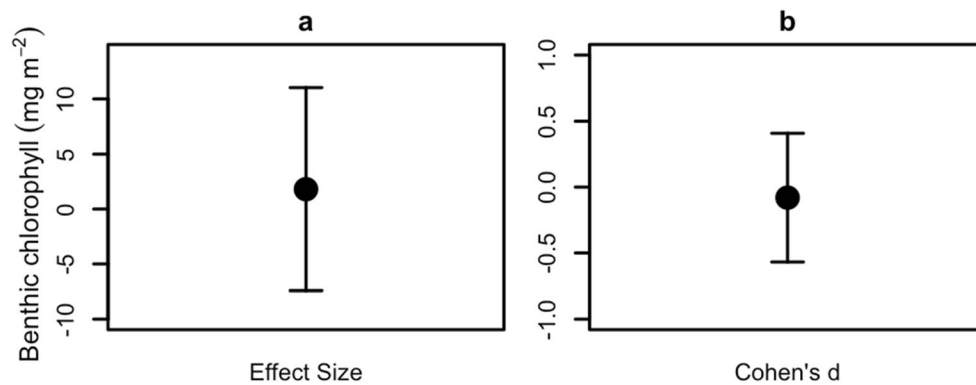


Fig. 3 **a** Mean effect size of macroalgal mats on benthic chlorophyll (mg m^{-2}) and **b** Cohen's $d \pm 95\%$ confidence intervals (C.I.) from the meta-analysis. Cohen's d is effect size normalized to standard deviation. Eight studies were included and each tested multiple treatments; $n = 37$ for effect size and $n = 35$ for Cohen's d (two treatments did not report

standard deviation). Note the different scales for the two plots. Effect size was positive, and Cohen's d was negative. In both cases, C.I. overlapped zero and are not considered statistically significant. The variability was very high in effect size and Cohen's d

similar to those we measured in the comparative survey (see below, Supplemental Material, Table 1A). The mean effect size was 1.8 mg m^{-2} (± 9.2 95% C.I.), and the effect sizes in MPB ranged between -76 and 75 mg m^{-2} (Fig. 3). The mean Cohen's d was -0.1 (± 0.5 95% C.I.) with values ranging between -5.1 and 3.8 (Fig. 3). In both formulations, the mean effect size was not different from 0, and effect sizes and Cohen's d indices were spread across positive and negative values (Fig. 3).

MPB may increase chlorophyll concentrations as a compensatory response to low-light availability. This process in diatoms is overcome at very low light (García-Robledo et al. 2012). Therefore, it is possible that when intermediate amounts of macroalgae are found shading MPB (between 40 and 100 g dw m^{-2}), chlorophyll becomes a poor proxy for biomass. At high macroalgal biomasses, diatoms presumably cease producing high concentrations of chlorophyll because photosynthesis is inhibited (García-Robledo et al. 2012). In order to account for the possibility of compensatory chlorophyll increases at intermediate macroalgal biomass, treatments with biomass between 40 and 100 g dw m^{-2} were omitted and effect sizes recalculated. This resulted in a Cohen's d that was small and positive and not different from zero (0.2 ± 0.4 95% C.I.). Likewise, the mean effect size was non-significant and positive ($2.9 \text{ mg m}^{-2} \pm 10.8$ 95% C.I.). Finally, no trend was found between effect size or Cohen's d and the mass of macroalgae used in the experiment (Supplemental Material, Fig. 2A).

Macroalgal Survey

Macroalgal biomass for vegetated mudflats in the VCR varied between 8.8 and 39.7 g dw m^{-2} (Fig. 4). One mudflat studied remained unvegetated throughout the summer (and has remained unvegetated in subsequent years of observation) (unpublished data). Mean macroalgal biomass for mudflats in the

VCR was $17.7 (\pm 4.0 \text{ SE}) \text{ g dw m}^{-2}$ with *G. vermiculophylla* representing 63% of the biomass. The second most abundant macroalgal functional group was green foliose macroalgae belonging to the genus *Ulva*. Variance in macroalgal biomass was high; the mean coefficient of variation for total biomass among all sites was 0.68, and the coefficient of variation for *G. vermiculophylla* was 1.03. Percent cover of macroalgae ranged between 8 and 37% for vegetated mudflats. Fisherman's Island had the highest coverage (37%) probably because seagrass covered the mudflat entangling macroalgae. The mudflat with the second highest cover (Tom's Cove, 31%) may represent a more accurate maximum cover value for typical flats that lack rooted vegetation.

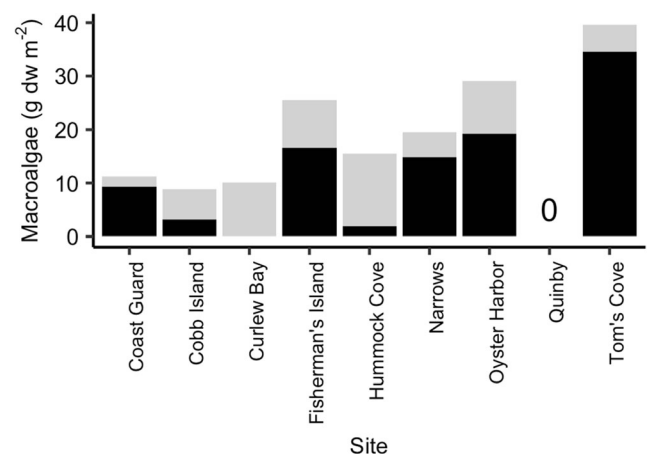


Fig. 4 Bars show mean biomass of all macroalgae from nine mudflats sampled in May and July 2015. All sites were sampled in both months except for the Coast Guard site, which was only sampled in July. Mean *Gracilaria vermiculophylla* (black) and all non-*Gracilaria vermiculophylla* macroalgae (gray) are displayed. Quinby was unvegetated (mean biomass = 0) across all sampling events. Average coefficient of variation for *Gracilaria vermiculophylla* among all sites was 1.03 and average coefficient of variation for total macroalgae across all sites was 0.68

Comparative Survey

Macroalgal patch biomass collected in September 2017 from the comparative survey locations spanned an order of magnitude, ranging between 37.1 and 360.2 g dw m⁻², with *G. vermiculophylla* representing 99% of the biomass and *Ulva* spp. constituting the rest (Fig. 5). Benthic chlorophyll concentrations ranged significantly as well, between 37.5 (± 9.0 SE) and 126.5 (± 16.2 SE) mg m⁻², and the mean concentration across all samples was 70.5 mg m⁻² (± 4.3 SE). There was no significant relationship between macroalgae and microalgae, and macroalgal biomass explained little of the variation ($R^2 = 0.02$) (Fig. 5). The mean benthic chlorophyll concentration was similar no matter how thick the macroalgal mat overgrowing the mudflat surface (Fig. 5).

Macroalga-Removal Manipulation

The mudflats in Oyster Harbor were vegetated with relatively high macroalgal biomass and somewhat more *G. vermiculophylla* than the average percent for the VCR. In the 2015 survey, biomass at this site was 29.1 (± 0.73 SE) g dw m⁻² of macroalgae with *Gracilaria vermiculophylla* comprising 66% of the macroalgal biomass. In 2016 at the beginning of the manipulation, the control plot was vegetated over 30% of the mudflat and the treatment plot had macroalgae over 24% of the plot area. Biomasses for the control and treatment sites at the beginning of the experiment were 42 and 33 g dw m⁻², respectively. The removal reduced cover to 11% and mass to 13 g dw m⁻² on the treatment site.

The benthic chlorophyll results suggest that macroalgae at typical quantities in the VCR have a neutral net effect on MPB. Benthic chlorophyll concentrations varied widely

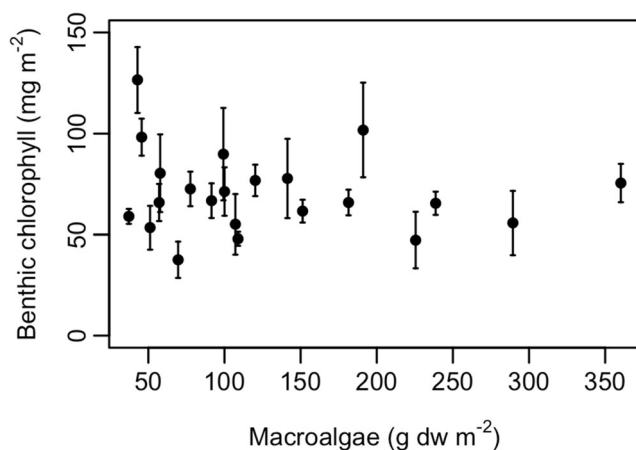


Fig. 5 Bi-plot of macroalgal biomass in grams dry weight per square meter against mean benthic chlorophyll concentrations in milligrams per square meter ± SE ($n = 3$) from the comparative survey conducted in the VCR. There was no correlation between the two variables

during the experiment from 27 to 131 mg m⁻². Benthic chlorophyll concentration was similar in all groups, and the interaction between time period (before or after treatment) and the treatment (control or removal plot) was not significant (Table 2; Fig. 6). The naturally bare control plot observed in the second half of the experiment corroborates the neutral effect of the macroalgal mat, as MPB is similar to the means of the experimental groups (Fig. 6).

Discussion

We did not find convincing evidence that MPB biomass is consistently, negatively impacted by macroalgae. The response of MPB was variable in the meta-analysis, comparative survey, and manipulation. The mean effect size for MPB in the meta-analysis was not different from zero, and the range was 150 mg m⁻². Many treatments had an increase in MPB when macroalgal biomass was added as compared to control. The comparative survey found no trend in MPB across a large range of macroalgal biomass. And in the large-scale manipulation, the removal of macroalgae did not produce a change in the MPB biomass as compared to a control plot covered in macroalgae.

Study Designs and Conditions

Taking together the meta-analysis, comparative survey, and manipulation, there was no trend in microalgal biomass along a wide range of tested macroalgal biomasses and functional forms. Macroalgal biomass spanned two orders of magnitude across these three study components. Excluding the bare-sediment control treatments (0 g dw m⁻²), macroalgal mass varied between 13 g dw m⁻² (this study) and 1300 g dw m⁻² (Green and Fong 2016). Average

Table 2 Results from the two-way ANOVA with an interaction used to test for the impact of the large-scale macroalga-removal on MPB biomass (using benthic chlorophyll as a proxy)

Source	DF	SS	MS	F	P
Treatment	1	0.028	0.028	1.072	0.312
Time	1	0.074	0.074	2.811	0.109
Treatment × time	1	0.000	0.000	0.000	0.983
Residuals	20	0.527	0.264		

Control and treatment plots (each 70 × 50 m) were observed in July and August for 12 days prior to removal of macroalgae from the treatment plot and 18 days after. If the removal had been significant, we would have expected to see a significant interaction effect between treatment and time (treatment × time in table below). The interaction was not significant, indicating that the removal of macroalgae did not significantly affect the biomass of MPB

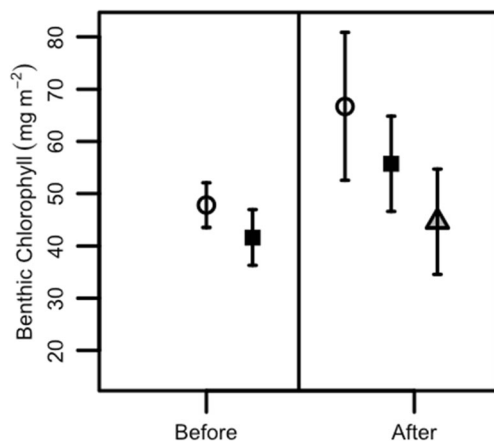


Fig. 6 Benthic chlorophyll concentrations before and after macroalgal mat removal; vertical line separates the before macroalgal removal period from the after removal period. Values are mean $\pm$ 1 SE ($n = 6$). Treatments include a control plot covered in macroalgae (open circle), an experimental plot where macroalgae was removed (black square), and a bare control mudflat sampled only in the second half of the experiment (gray triangle). The naturally bare mudflat was for comparative purposes only and not evaluated statistically

macroalgal biomass in the VCR was similar to amounts found previously in this system (Thomsen et al. 2006). Samples collected for the comparative survey in the bays surrounding Oyster Harbor, the Narrows, and the Coast Guard site had a higher proportion of *G. vermiculophylla* than average for the VCR. This might have been a result of the sites proximity to the mainland, and because samples were collected in September and other more ephemeral macroalgae may have senesced (Thomsen et al. 2006). Macroalgal taxa in the meta-analysis included red filamentous, green filamentous, and green foliose functional-form groups. There were mostly *Gracilaria* spp. and *Ulva* spp. Testing different functional forms is important since recalcitrance and longevity vary among groups, which can have consequences for ecosystem nutrient dynamics and metabolism (McGlathery et al. 2001, 2007, 2012).

Study designs included in the original work presented here and the meta-analysis involved surveys, field experiments, core-incubations, and mesocosm studies. These methods provided varying degrees of control and realism; effects are often easier to identify in controlled micro- or mesocosm experiments but lack the context and interactions found in situ (Carpenter 1996). Negative impacts of macroalgae on MPB biomass occurred in some mesocosm experiments (Sundbäck et al. 1990; García-Robledo and Corzo 2011) (Table 1). However, the negative impacts observed in those mesocosm studies have not been replicated in field-based manipulations. In situ field experiments produced neutral results, or found positive effects of macroalgal mass on MPB (Corzo et al. 2009; Bohórquez et al. 2013; Green and Fong 2016, this study). Our work

demonstrates the importance of in situ research for assessing outcomes and ecosystem consequences.

For all of the studies considered here, experiments and surveys conducted at meso- and large scales ($\geq 1 \text{ m}^2$) did not show a significant relationship between macroalgae and MPB, and the responses of MPB in experiments that were smaller than 1 m^2 were variable. It is important to consider scale in experiments that measure MPB (Weerman et al. 2010; Pratt et al. 2015). For example, sediment grain size, hydrodynamic forces, and faunal grazing may all affect MPB at different temporal and spatial scales, such that patchy distributions and stochastic growth rates could conceal a net effect of macroalgae on MPB (Weerman et al. 2010; Mariotti and Fagherazzi 2012; Pratt et al. 2015). Likewise, a significant relationship detected at a small scale may not occur at an ecosystem scale over hundreds or thousands of square meters. Our approach addressed these problems by evaluating results from multiple studies encompassing different spatial scales. Temporal scale was not a concern for this work as all study designs in the original work, and all but two treatments in the meta-analysis, integrated over longer time periods than necessary for MPB to recover from a disturbance (Pace et al. 1979; Rossi et al. 2007; Leite et al. 2012). The spatial scales of field, microcosm, and mesocosm experiments ranged between a few hundred square centimeters and more than 3000 m^2 , and the length of observations ranged between a single sampling event and treatments lasting 8 weeks (Corzo et al. 2009; Bohórquez et al. 2013; Green and Fong 2016; this study). As mentioned above, there were a few instances where significant negative results from macroalgal incubation on MPB biomass were found in tanks or cores; however, measurements were made on a relatively small scale (hundreds of cm) (Sundbäck et al. 1990; García-Robledo and Corzo 2011).

Microalgal vertical migration occurs according to tidal and solar cycles, and we attempted to control for these movements. In the large-scale manipulation, we standardized sampling to mid-tide during the daytime so both the tide and light conditions would be similar. We sampled to a 1-cm depth in the large-scale manipulation and to 2-cm depth in the comparative survey; these methods did not produce significantly different concentrations of chlorophyll. As the tide rises, microalgae migrate downward away from the sediment surface to avoid being dislodged by wave action (Kingston 1999). However, during high tide in muddy sediments, the bulk of MPB is still in the top centimeters of sediment, so migration should not have presented a problem for our sampling methods (Du et al. 2010). Additionally, it is possible that MPB migrate less beneath macroalgal mats than MPB on bare substrate because macroalgal mats can decrease sediment erosion

(Lawson et al. 2012; Venier et al. 2012). Variability in MPB was not likely affected by vertical migration in the comparative survey or large-scale manipulation presented here, despite sampling to a relatively shallow depth.

Macroalgal-MPB Competition or Facilitation

Collectively, the results support the conclusion that macroalgal canopies do not usually have a negative impact on microphytobenthos. This suggests an even more fundamental conclusion: in shallow coastal systems, MPB and macroalgal mats do not compete for critical resources such as light and nutrients. While competition for these resources was not explicitly tested in the meta-analysis, comparative survey, or manipulation, there is evidence to support this claim from the literature. Co-existence of MPB and macroalgal canopies may occur because resources are abundant, because these organisms possess adaptations that allow shared, limited resources to be partitioned or as a result of facilitation.

Considering competition for light, it is possible that both adaptations and sufficient light resources resolved any possible contest. Light attenuates exponentially through macroalgal mats (Krause-Jensen et al. 1996). Macroalgal mats in the VCR are on average 2–3-cm thick; using the relationship developed in Krause-Jensen et al., mats should reduce irradiance by ~50% (1996, pers. obs.). If one assumes an average photosynthetic photon flux of ~400 $\mu\text{mol m}^{-2} \text{s}^{-1}$ at the top of the macroalgal mat at mid-tide (0.75–1.5-m water depth) in the VCR, then there is a photon flux of ~200 $\mu\text{mol m}^{-2} \text{s}^{-1}$ at the sediment surface beneath the macroalgal canopy (Table 3). This quantity of light is still associated with significant primary production by algae, which can continue photosynthesis with as little as 5–10 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$; therefore, competition for

light may not be occurring beneath most macroalgal mats because there is sufficient light availability for MPB (Krause-Jensen et al. 1996, Cahoon and references therein 1999; Pace and Lovett 2013). However, sediments beneath macroalgal mats of much greater than 2-cm depth can still contain high quantities of chlorophyll (Green and Fong 2016, this study). In the comparative survey of macroalgal biomass and MPB, the densest macroalgal mat sampled was 360 g dw m^{-2} and at least 5-cm thick, yet there was $75.5 \pm 9.5 \text{ mg m}^{-2}$ chlorophyll in the sediment below the mat. Light below this depth of algae should be <10% of incident irradiance at the surface of the macroalgal canopy. Still, 75.5 mg m^{-2} chlorophyll nearly equals the third quartile of the MPB data from the comparative survey ($Q3 = 77.5 \text{ mg m}^{-2}$). Counterintuitively, MPB biomass was higher than the average and median concentrations found throughout the comparative survey beneath the deepest and heaviest macroalgal mat sampled.

That the MPB were high beneath very dense macroalgal mats suggests some facilitation, or at least compensation, of MPB by macroalgae via leaking nutrients and carbon. Diatoms are capable of heterotrophy, and experiments have demonstrated the rate of uptake and oxidation of organic compounds increases significantly in the dark (White 1974; Tuchman et al. 2006). It is likely that macroalgal mats provide a significant amount of labile organic carbon to benthic microalgae from exudates and degrading thalli (Sundbäck and McGlathery 2005). This carbon subsidy may allow MPB to compensate for light limitation beneath very thick canopies with heterotrophic metabolism of macroalgal carbon. Similarly, nutrient limitation is unlikely induced for MPB by macroalgal canopies because macroalgae release significant amounts of dissolved nitrogen back into the water column. Macroalgae rapidly take up nutrients but have low assimilation efficiencies (Sundbäck and McGlathery 2005; Tyler and McGlathery 2006). Further evidence of MPB facilitation by macroalgae comes from an experiment using isotopic nitrogen tracers; *Gracilaria vermiculophylla* enriched with nitrogen-15 transferred nitrogen to benthic microalgae (Gulbrandsen and McGlathery 2013). Nutrient and carbon subsidies from macroalgae to MPB together may compensate for photosynthetic limitation from macroalgal shading.

One other mechanism of MPB facilitation by macroalgae could be protection from UV radiation. UV-B radiation can negatively impact diatom photosynthetic activity and biomass, and diatoms will migrate downward to avoid intense UVB radiation (Underwood et al. 1999). Macroalgal mats filter both photosynthetically active radiation (PAR) as well as UV-A and UV-B radiation (Krause-Jensen et al. 1996; Sundbäck et al. 1996). Light attenuation from macroalgal mats may present a trade-off for MPB, whereby the negative impacts of reduced PAR are counteracted by the positive effects of reduced UV-B radiation.

Table 3 Environmental variables collected during the large-scale manipulation from the mudflat ecosystem under study in Oyster Harbor

Variable	Units	Number	Mean $\pm$ SD
Temperature	C°	30	29.8 $\pm$ 3.4
Salinity	ppt	5	32.2 $\pm$ 0.3
Light intensity	$\mu\text{mol m}^{-2} \text{s}^{-1}$	30	395 $\pm$ 726
TSS	mg L^{-1}	12	34.2 $\pm$ 10.4
SOM	%	116	3.7 $\pm$ 0.7
Porosity	%	112	63.8 $\pm$ 10.2

Number, sample size, is a combination of spatial and temporal replicates. Means and SD are presented for each variable. Porosity is the percentage of pore space in the sediment by volume. Light intensity was measured at mid-tide 20 cm above the sediment, which was just above the macroalgal mat

TSS total suspended solids for the water above the mudflats, SOM sediment organic matter

Generality of Results

The generality of the field results presented here is partly dependent on how similar the VCR is to other coastal systems. Two features of the VCR are pertinent. First, the VCR is a low-nutrient system and may not be representative of macroalgal biomass found in more heavily impacted estuaries (McGlathery et al. 2007). Second, macroalgal biomass is dominated by a single, non-native species in the VCR (Thomsen et al. 2005, 2006) (Fig. 4). We believe that the first issue is scale-dependent. Our results may be comparable at small scales (m^2) because macroalgal biomass can reach high biomass in patches, but possibly not at larger scales (hectares), because macroalgal biomass is lower at this scale and spatially integrated results may differ. For the second problem, we considered how *G. vermiculophylla* could affect MPB, but we did not observe any ecosystem behaviors that indicated a unique effect from a non-native in the system (Ramus et al. 2017). Both of these arguments are expanded below to support our position that our results are generalizable to other estuaries.

Average macroalgal biomass at a mudflat-scale in the VCR (18 g dw m^{-2}) is low compared to biomasses reported for intertidal flats in some other estuaries (e.g., 45–180 g dw m^{-2} in Waquoit Bay, MA) and is well below reported threshold levels for severe negative impacts (175 g dw m^{-2}) (Teichberg et al. 2008; Sutula et al. 2014). The integrated MPB response in our large-scale manipulation is not likely comparable to a mudflat with 10× higher average macroalgal biomass. However, at small sampling scales ($< 1 \text{ m}^2$), our results are comparable. The maximum macroalgal biomass collected in the comparative survey was $> 350 \text{ g dw m}^{-2}$, and previous research in the VCR measured macroalgal biomass up to 650 g dw m^{-2} (McGlathery et al. 2001). These biomasses are similar to those reported for other systems, and as discussed previously, MPB was quite high beneath the heaviest macroalgal patch sampled. Macroalgal biomass is lower at large-scales in the VCR than in other estuaries; however, it is patchily distributed and the neutral response of MPB beneath high biomass patches did not significantly differ from the pattern observed in the large-scale manipulation. Nonetheless, further study of MPB-macroalgal interactions is warranted for more nutrient-enriched estuaries given the relatively oligotrophic conditions of the VCR.

Non-native organisms often alter ecosystem function, making comparison with un-invaded systems problematic (Strayer 2012). In the VCR, the dominant non-native *G. vermiculophylla* may actually change some processes by acting as a foundation species that reduces current speed and wave energy and increases epifaunal diversity and abundance relative to bare mudflats (Thomsen et al. 2010; Byers et al. 2012; Wright et al. 2014; Ramus et al. 2017).

Increased epifaunal abundance could result in lower MPB biomass due to higher grazing pressure. Since we did not see a decrease in MPB below macroalgal mats, this effect should not have confounded our results. Marine foundation species increase MPB by attenuating hydrodynamic stresses and improving sediment stability (Donadi et al. 2013). Accordingly, in the VCR, it could be possible that any negative effects to MPB biomass induced by competition with *G. vermiculophylla* are counteracted by protection from hydrodynamic stress. However, velocities measured on a mudflat adjacent to the manipulation study site were extremely low ($< 1 \text{ cm s}^{-1}$), reducing the likelihood of the protective-*G. vermiculophylla* hypothesis explaining the macroalga-removal results (unpublished data). The similarity in MPB biomass patterns beneath *G. vermiculophylla* and foliose and filamentous *Ulva* spp. indicates that these results are not species-specific (Corzo et al. 2009; Bohórquez et al. 2013; Hardison et al. 2013; Green and Fong 2016, this study). MPB appear to be resilient to the stresses induced by macroalgae regardless of whether the species is native or invasive.

Conclusions

In summary, macroalgae do not diminish MPB biomass across a wide range of conditions in the VCR and in many prior experimental studies. The impacts of macroalgae are context dependent. High macroalgal biomasses resulting from eutrophication and species invasions can have significant negative impacts on seagrass meadows, coral reefs, water quality, and other features (McGlathery et al. 2007; Williams and Smith 2007; Williams et al. 2013). Low to moderately high macroalgal biomass may facilitate a variety of processes that increase structural and biological complexity in coastal ecosystems (Williams and Smith 2007; Thomsen et al. 2010; Byers et al. 2012; Ramus et al. 2017). The findings of this study support a generalization that MPB biomass is not negatively impacted by high macroalgal biomass. One focus for future work is determining how macroalgal impacts vary among coastal ecosystems. A second focus is determining the conditions that promote the transition of macroalgae to nuisance levels.

Acknowledgements We are especially thankful to Nicholas Braun for assistance with macroalgal surveys; to Lillian Aoki, Amelie Berger, Martin Volaric, and Mo Tatlehego for assistance in the macroalgal removal manipulation; to Joseph Siegel for field support; to David Boyd, David Lee, Tripp Byrd, Donna Fauber, and other Anheuser-Busch Coastal Research Center staff for logistical support; and to Karen McGlathery, Meg Miller, and Arthur Schwarzschild for guidance on study design, methods, and lab work. We are grateful for the comments and suggestions made by reviewers that improved this manuscript. The Nature Conservancy and The National Park Service provided access to sites.

References

- Bohórquez, J., S. Papaspyrou, M. Yúfera, S.A. van Bergeijk, E. García-Robledo, J.L. Jiménez-arias, M. Bright, and A. Corzo. 2013. Effects of green macroalgal blooms on the meiofauna community structure in the Bay of Cádiz. *Marine Pollution Bulletin* 70 (1-2): 10–17.
- Byers, J.E., P.E. Gribben, C. Yeager, and E.E. Sotka. 2012. Impacts of an abundant introduced ecosystem engineer within mudflats of the southeastern US coast. *Biological Invasions* 14 (12): 2587–2600.
- Cahoon, L.B. 1999. The role of benthic microalgae in neritic ecosystems. *Oceanography and Marine Biology: An Annual Review* 37: 47–86.
- Carpenter, S.R. 1996. Microcosm experiments have limited relevance for community and ecosystem ecology. *Ecology* 77 (3): 677–680.
- Carpentier, A., S. Como, C. Dupuy, C. Lefrançois, and E. Feunteun. 2014. Feeding ecology of *Liza* spp. in a tidal flat: evidence of the importance of primary production (biofilm) and associated meiofauna. *Journal of Sea Research* 92: 86–91.
- Corzo, A., S.A. van Bergeijk, and E. García-Robledo. 2009. Effects of green macroalgal blooms on intertidal sediments: net metabolism and carbon and nitrogen contents. *Marine Ecology Progress Series* 380: 81–93.
- Currin, C.A., L.A. Levin, T.S. Talley, R. Michener, and D. Talley. 2011. The role of cyanobacteria in Southern California salt marsh food webs. *Marine Ecology* 32 (3): 346–363.
- Donadi, S., J. Westra, E.J. Weerman, T. van der Heide, E.M. van der Zee, J. van de Koppel, H. Olf, T. Piersma, H.W. van der Veer, and B.K. Eriksson. 2013. Non-trophic interactions control benthic producers on intertidal flats. *Ecosystems* 16 (7): 1325–1335.
- Douglass, J.G., J.E. Duffy, and E.A. Canuel. 2011. Food web structure in a Chesapeake Bay eelgrass bed as determined through gut contents and ¹³C and ¹⁵N isotope analysis. *Estuaries and Coasts* 34 (4): 701–711.
- Du, G.Y., M. Son, S. An, and I.K. Chung. 2010. Temporal variation in the vertical distribution of microphytobenthos in intertidal flats of the Nakdong River estuary, Korea. *Estuarine, Coastal and Shelf Science* 86 (1): 62–70.
- García-Robledo, E., and A. Corzo. 2011. Effects of macroalgal blooms on carbon and nitrogen biogeochemical cycling in photoautotrophic sediments: an experimental mesocosm. *Marine Pollution Bulletin* 62 (7): 1550–1556.
- García-Robledo, E., A. Corzo, S. Papaspyrou, and E.P. Morris. 2012. Photosynthetic activity and community shifts of microphytobenthos covered by green macroalgae. *Environmental Microbiology Reports* 4 (3): 316–325.
- Green, L., and P. Fong. 2016. The good, the bad and the *Ulva*: the density dependent role of macroalgal subsidies in influencing diversity and trophic structure of an estuarine community. *Oikos* 125 (7): 988–1000.
- Grinham, A.R., T.J.B. Carruthers, P.L. Fisher, J.W. Udy, and W.C. Dennison. 2007. Accurately measuring the abundance of benthic microalgae in spatially variable habitats. *Limnology and Oceanography: Methods* 5: 119–125.
- Gulbransen, D., and K.J. McGlathery. 2013. Nitrogen transfers mediated by a perennial, non-native macroalga: a ¹⁵N tracer study. *Marine Ecology Progress Series* 482: 299–304.
- Gulbransen, D.J., K.J. McGlathery, M. Marklund, J.N. Norris, C. Frederico, and D. Gurgel. 2012. *Gracilaria vermiculophylla* (Rhodophyta, Gracilariales) in the Virginia coastal bays, USA: COX1 analysis reveals high genetic richness of an introduced macroalga. *Journal of Phycology* 48 (5): 1278–1283.
- Hardison, A.K., E.A. Canuel, I.C. Anderson, and B. Veuger. 2010. Fate of macroalgae in benthic systems: carbon and nitrogen cycling within the microbial community. *Marine Ecology Progress Series* 414: 41–55.
- Hardison, A.K., I.C. Anderson, E.A. Canuel, C.R. Tobias, and B. Veuger. 2011. Carbon and nitrogen dynamics in shallow photic systems: interactions between macroalgae, microalgae, and bacteria. *Limnology and Oceanography* 56 (4): 1489–1503.
- Hardison, A.K., E.A. Canuel, I.C. Anderson, C.R. Tobias, B. Veuger, and M.N. Waters. 2013. Microphytobenthos and benthic macroalgae determine sediment organic matter composition in shallow photic sediments. *Biogeosciences* 10 (8): 5571–5588.
- Kennett, D.M., and P.E. Hargraves. 1985. Benthic diatoms and sulfide fluctuations: upper basin of Pettaquamscutt River, Rhode Island. *Estuarine, Coastal and Shelf Science* 21 (4): 577–586.
- Kingston, M.B. 1999. Wave effects on the vertical migration of two benthic microalgae: *Hantzschia virgata* var. *intermedia* and *Euglena proxima*. *Estuaries* 22 (1): 81–91.
- Kollars, N.M., J.E. Byers, and E.E. Sotka. 2016. Invasive décor: an association between a native decorator worm and a non-native seaweed can be mutualistic. *Marine Ecology Progress Series* 545: 135–145.
- Krause-Jensen, D., K.J. McGlathery, S. Rysgaard, and P.B. Christensen. 1996. Production within dense mats of filamentous macroalga *Chaetomorpha linum* in relation to light and nutrient availability. *Marine Ecology Progress Series* 134: 207–216.
- Kuwae, T., E. Miyoshi, S. Hosokawa, K. Ichimi, J. Hosoya, T. Amano, T. Moriya, M. Kondoh, R.C. Ydenberg, and R.W. Elner. 2012. Variable and complex food web structures revealed by exploring missing trophic links between birds and biofilm. *Ecology Letters* 15 (4): 347–356.
- Lawson, S.E., K.J. McGlathery, and P.L. Wiberg. 2012. Enhancement of sediment suspension and nutrient flux by benthic macrophytes at low biomass. *Marine Ecology Progress Series* 448: 259–270.
- Lee, D.K. 2016. Alternatives to p value: confidence interval and effect size. *Korean Journal of Anesthesiology* 69 (6): 555–562.
- Leite, L.G., Á.M. Ciotti, and R.A. Christofolletti. 2012. Abundance of biofilm on intertidal rocky shores: can trampling by humans be a negative influence? *Marine Environmental Research* 79: 111–115.
- Lorenzen, C.J. 1967. Determination of chlorophyll and phaeo-pigments: spectrophotometric equations. *Limnology and Oceanography* 12 (2): 343–346.
- Lyons, D.A., C. Arvanitidis, A.J. Blight, E. Chatzinikolaou, T. Guy-Haim, J. Kottla, H. Orav-Kotta, A.M. Queirós, G. Rilov, P.J. Somerfield, and T.P. Crowe. 2014. Macroalgal blooms alter community structure and primary productivity in marine ecosystems. *Global Change Biology* 20 (9): 2712–2724.
- Mariotti, G., and S. Fagherazzi. 2012. Modeling the effect of tides and waves on benthic biofilms. *Journal of Geophysical Research* 117: 1–14.
- McGlathery, K.J., I.C. Anderson, and A.C. Tyler. 2001. Magnitude and variability of benthic and pelagic metabolism in a temperate coastal lagoon. *Marine Ecology Progress Series* 216: 1–15.
- McGlathery, K.J., K. Sundbäck, and I.C. Anderson. 2007. Eutrophication in shallow coastal bays and lagoons: the role of plants in the coastal filter. *Marine Ecology Progress Series* 348: 1–18.
- McGlathery, K.J., K. Sundbäck, and P. Fong. 2012. Estuarine benthic algae. In *Estuarine ecology*, ed. J.W. Day, B.C. Crump, W.M. Kemp, and A. Yanez-Arancibia 2nd, 203–234. Hoboken: Wiley.
- Nilsson, P., B. Jönsson, I.L. Swanberg, and K. Sundbäck. 1991. Response of marine shallow-water sediment system to increased load of inorganic nutrients. *Marine Ecology Progress Series* 71: 275–290.
- Pace, M.L., and G.M. Lovett. 2013. Primary production: the foundation of ecosystems. In *Fundamentals of ecosystem science*, ed. K.C. Weathers, D.L. Strayer, and G.E. Likens, 27–51. San Diego: Elsevier Science Publishing Co.
- Pace, M.L., S. Shimmel, and W.M. Darley. 1979. The effect of grazing by a gastropod, *Nassarius obsoletus*, on the benthic microbial community of a salt marsh mudflat. *Estuarine and Coastal Marine Science* 9 (2): 121–134.
- Pratt, D.R., C.A. Pilditch, A.M. Lohrer, S.F. Thrush, and C. Kraan. 2015. Spatial distributions of grazing activity and microphytobenthos reveal scale-dependent relationships across a sedimentary gradient. *Estuaries and Coasts* 38 (3): 722–734.

- Ramus, A.P., B.R. Silliman, M.S. Thomsen, and Z.T. Long. 2017. An invasive foundation species enhances multifunctionality in a coastal ecosystem. *Proceedings of the National Academy of Sciences* 114 (32): 8580–8585.
- Rossi, F., R.M. Forster, F. Montserrat, M. Ponti, A. Terlizzi, T. Ysebaert, and J.J. Middelburg. 2007. Human trampling as short-term disturbance on intertidal mudflats: effects on macrofauna biodiversity and population dynamics of bivalves. *Marine Biology* 151 (6): 2077–2090.
- Saint-Béat, B., C. Dupuy, P. Bocher, J. Chalumeau, M. De Crignis, C. Fontaine, K. Guizien, J. Lavaud, S. Lefebvre, H. Montanié, J.L. Mouget, F. Orvain, P.Y. Pascal, G. Quaintenne, G. Radenac, P. Richard, F. Robin, A.F. Vézina, and N. Niquil. 2013. Key features of intertidal food webs that support migratory shorebirds. *PLoS One* 8: 1–18.
- Smith, E.P. 2002. BACI design. In *Encyclopedia of environmetrics*, ed. A.H. El-Shaarawi and W.W. Piegorsch, vol. 1, 141–148. Chichester: Wiley.
- Strayer, D.L. 2012. Eight questions about invasions and ecosystem functioning. *Ecology Letters* 15 (10): 1199–1210.
- Sundbäck, K., and K.J. McGlathery. 2005. Interactions between benthic macroalgal and microalgal mats. In *Interactions between macro- and microorganisms in marine sediments*, ed. E. Kristensen, R.R. Haese, and J.E. Kostka, 7–29. Washington, DC: American Geophysical Union.
- Sundbäck, K., B. Jönsson, P. Nilsson, and I. Lindström. 1990. Impact of accumulating drifting macroalgae on a shallow-water sediment system: an experimental study. *Marine Ecology Progress Series* 58: 261–274.
- Sundbäck, K., L. Carlson, C. Nilsson, B. Jönsson, A. Wulff, and S. Odmark. 1996. Response of benthic microbial mats to drifting green algal mats. *Aquatic Microbial Ecology* 10: 195–208.
- Sutula, M., L. Green, G. Cicchetti, N. Detenbeck, and P. Fong. 2014. Thresholds of adverse effects of macroalgal abundance and sediment organic matter on benthic habitat quality in estuarine intertidal flats. *Estuaries and Coasts* 37 (6): 1532–1548.
- Teichberg, M., S.E. Fox, C. Aguila, Y.S. Olsen, and I. Valiela. 2008. Macroalgal responses to experimental nutrient enrichment in shallow coastal waters: growth, internal nutrient pools, and isotopic signatures. *Marine Ecology Progress Series* 368: 117–126.
- Thimijan, R.W., and R.D. Heins. 1983. Photometric, radiometric, and quantum light units of measure: a review of procedures for interconversion. *Hortscience* 18: 818–822.
- Thomsen, M.S., and K.J. McGlathery. 2005. Facilitation of macroalgae by the sedimentary tube forming polychaete *Diopatra cuprea*. *Estuarine, Coastal and Shelf Science* 62 (1–2): 63–73.
- Thomsen, M.S., and K.J. McGlathery. 2007. Stress tolerance of the invasive macroalgae *Codium fragile* and *Gracilaria vermiculophylla* in a soft-bottom turbid lagoon. *Biological Invasions* 9 (5): 499–513.
- Thomsen, M.S., C.F.D. Gurgel, S. Fredericq, and K.J. McGlathery. 2005. *Gracilaria vermiculophylla* (Rhodophyta, Gracilariales) in Hog Island Bay, Virginia: a cryptic alien and invasive macroalga and taxonomic correction. *Journal of Phycology* 42: 139–141.
- Thomsen, M.S., K.J. McGlathery, and A.C. Tyler. 2006. Macroalgal distribution patterns in a shallow, soft-bottom lagoon, with emphasis on the nonnative *Gracilaria vermiculophylla* and *Codium fragile*. *Estuaries and Coasts* 29 (3): 465–473.
- Thomsen, M.S., K.J. McGlathery, A. Schwarzschild, and B.R. Silliman. 2009. Distribution and ecological role of the non-native macroalga *Gracilaria vermiculophylla* in Virginia salt marshes. *Biological Invasions* 11 (10): 2303–2316.
- Thomsen, M.S., T. Wernberg, A. Altieri, F. Tuya, D. Gulbransen, K.J. McGlathery, M. Holmer, and B.R. Silliman. 2010. Habitat cascades: the conceptual context and global relevance of facilitation cascades via habitat formation and modification. *Integrative and Comparative Biology* 50 (2): 158–175.
- Tuchman, N.C., M.A. Schollett, S.T. Rier, and P. Geddes. 2006. Differential heterotrophic utilization of organic compounds by diatoms and bacteria under light and dark conditions. *Hydrobiologia* 561 (1): 167–177.
- Tyler, A.C., and K.J. McGlathery. 2006. Uptake and release of nitrogen by the macroalgae *Gracilaria vermiculophylla* (Rhodophyta). *Journal of Phycology* 42 (3): 515–525.
- Tyler, A.C., K.J. McGlathery, and I.C. Anderson. 2003. Benthic algae control sediment: water column fluxes of organic and inorganic nitrogen compounds in a temperate lagoon. *Limnology and Oceanography* 48 (6): 2125–2137.
- Underwood, G.J.C., and J. Kromkamp. 1999. Primary production by phytoplankton and microphytobenthos in estuaries. *Advances in Ecological Research* 29: 93–153.
- Valiela, I., J. McClelland, J. Hauxwell, P.J. Behr, D. Hersh, and K. Foreman. 1997. Macroalgal blooms in shallow estuaries: controls and ecophysiological and ecosystem consequences. *Limnology and Oceanography* 42 (5part2): 1105–1118.
- Venier, C., J.F. da Silva, S.J. McLelland, R.W. Duck, and S. Lanzoni. 2012. Experimental investigation of the impact of macroalgal mats on flow dynamics and sediment stability in shallow tidal areas. *Estuarine, Coastal and Shelf Science* 112: 52–60.
- Weerman, E.J., J. van de Koppel, M.B. Eppinga, F. Montserrat, Q.X. Liu, and P.M.J. Herman. 2010. Spatial self-organization on intertidal mudflats through biophysical stress divergence. *The American Naturalist* 176 (1): E15–E32.
- White, A.W. 1974. Uptake of organic compounds by two facultatively heterotrophic marine centric diatoms. *Journal of Phycology* 10: 433–438.
- Williams, S.L., and J.E. Smith. 2007. A global review of the distribution, taxonomy, and impacts of introduced seaweeds. *Annual Review of Ecology, Evolution, and Systematics* 38 (1): 327–359.
- Williams, S.L., I.C. Davidson, J.R. Pasari, G.V. Ashton, J.T. Carlton, R.E. Crafton, R.E. Fontana, E.D. Grosholz, A. Whitman Miller, G.M. Ruiz, and C.J. Zabin. 2013. Managing multiple vectors for marine invasions in an increasingly connected world. *Bioscience* 63: 952–966.
- Wright, J.T., J.E. Byers, J.L. DeVore, and E.E. Sotka. 2014. Engineering or food? Mechanisms of facilitation by a habitat-forming invasive seaweed. *Ecology* 95 (10): 2699–2706.